

CM 6002 Digital Media and Information Technology



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INSTRUCTOR

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(Please Email/Facebook me first to make sure that I will be there.)

Course Syllabus and Lectures

For protecting our environment, all course syllabus and lectures can be downloaded online via the following links.

- ▣ Course Syllabus

- <http://gscm.nida.ac.th/~chutisant/CM6002/CS-CM6002.pdf>

- ▣ Lectures

- <http://gscm.nida.ac.th/~chutisant/CM6002/L1-CM6002.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM6002/L2-CM6002.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM6002/L3-CM6002.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM6002/L4-CM6002.pdf>

Evaluation (50%)



- Assignments, Participation and Attendance
10 %
- Project
40 %
(Technique-based, Creativity-based, and Research-based)

Class Schedule

๙	นวัตกรรมของสื่อดิจิทัลและเทคโนโลยีสารสนเทศในปัจจุบัน	๓	บรรยายเชิงอภิปราย
๑๐	สื่อดิจิทัลในรูปแบบเทคโนโลยีมีเดียสมัยใหม่: การรับรู้ทางสายตาและภาพลงตาสำหรับสื่อดิจิทัล	๓	บรรยายเชิงอภิปราย
๑๑	สื่อดิจิทัลในรูปแบบเทคโนโลยีมีเดียสมัยใหม่: การรับรู้ทางสายตาและภาพลงตาสำหรับสื่อดิจิทัล (ต่อ)	๓	บรรยายเชิงอภิปราย
๑๒	งานวิจัยของสื่อดิจิทัลและเทคโนโลยีสารสนเทศในปัจจุบัน	๓	บรรยายเชิงอภิปราย
๑๓	ปัญหาของสื่อดิจิทัลในแง่เทคโนโลยีสารสนเทศในปัจจุบัน	๓	บรรยายเชิงอภิปราย
๑๔	อนาคตของความเกี่ยวเนื่องกันของสื่อดิจิทัลและเทคโนโลยีสารสนเทศ	๓	บรรยายเชิงอภิปราย
๑๕	ทิศทางของการบูรณาการของสื่อดิจิทัลและเทคโนโลยีสารสนเทศ	๓	บรรยายเชิงอภิปราย

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Textbooks

- Augmented Reality: A Modern Vision-Based Approach, Author: Chutisant Kerdvibulvech, ISBN 9786167687216
- ชุตินันต์ เกิดวิบูลย์เวช. (2560) (2559) สื่อดิจิทัลใหม่..สื่อแห่งอนาคต. คณะนิเทศศาสตร์และนวัตกรรมการจัดการ สถาบันบัณฑิตพัฒนบริหารศาสตร์.

Academic Paper (Scopus)

- C. Kerdvibulvech, “A Methodology for Hand and Fingers Motion Analysis Using Adaptive Probabilistic Models,” EURASIP Journal on Embedded Systems, 9 pages, No. 18, Springer Publisher, 2014.

Websites and Links

- <http://ieeexplore.ieee.org/>
- <http://dl.acm.org/>
- <http://link.springer.com/bookseries/558>

- <http://www.dailynews.co.th/>
- <http://www.ecommerce-magazine.com/>
- <http://www.manager.co.th>

Overview

- ▣ Background
- ▣ From Digital Media to New Digital Media
- ▣ 3D Display / 3D Television
- ▣ Augmented Reality

Myself

- ▣ Received PhD in Computer Engineering from Keio University, Japan (2009)
- ▣ Assistant Professor (2015-2018), Associate Professor (2018-Now) at GSCM, **NIDA**
- ▣ Head of Department of Information and Communication Technology, Rangsit University (2009-2014)

Academic Work Experience

- Invited Lecturer and Research Committee in many universities in Thailand, such as
 - Chulalongkorn University Mahidol University
 - Sripatum University Kasetsart University
 - Panyapiwat Institute of Management

- Technical Committees for many academic international conferences and journals (Scopus and Scimago)

Media Work Experience

Newspapers

- IT Columnist for Dailynews newspaper, every Wednesday (page 23)



FROM DIGITAL MEDIA TO NEW DIGITAL MEDIA



Digital Media

Digital media are any media that are encoded in a machine-readable format. [*Technology Brief*. University of Guelph.]

1. Digital Imagery / Digital Video
2. Web Pages / Websites
3. Digital Audio, e.g. mp3s
4. e-books
5. Social Media



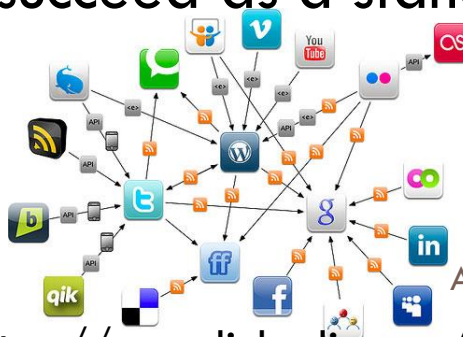
www.linkedin.com

Important Role of Digital Media

Online video has become ubiquitous, digital media consumption has become the norm.

Facts about Digital Media

1. Viewers are heavily engaged with digital media – less so with linear television and newspaper.
2. Television and newspaper still matter, but it can no longer succeed as a stand alone platform.



::Source::

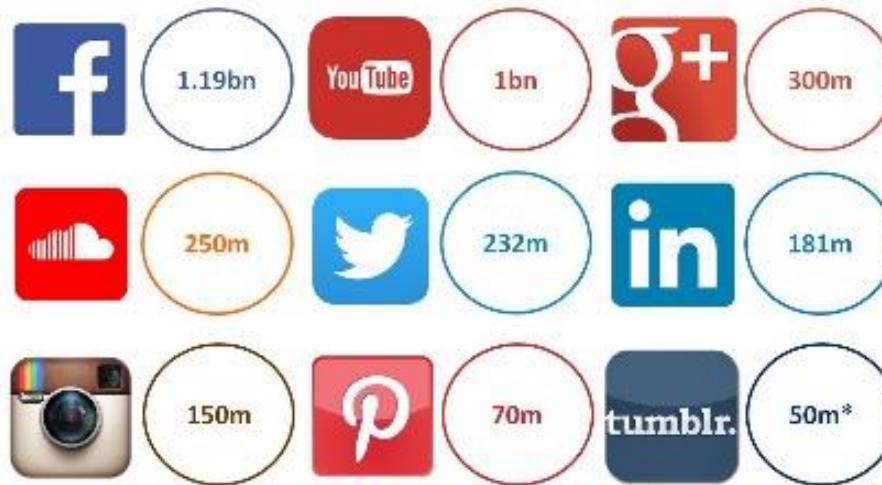
Digital Media Trends and the Future of Television, Dr. Seth Geiger, President SmithGeiger LLC (CA, USA)

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Trends for Digital Media for 2014

Active monthly users of the 'big 9'

Active Monthly Users of the 'Big 9'



As at 15 November 2013 (multiple sources cited)

*Tumblr does not release active monthly user data and this figure is based on an estimate from BusinessInsider.com. It does reveal that the site and its domains receives 300m visits per month.

kamber

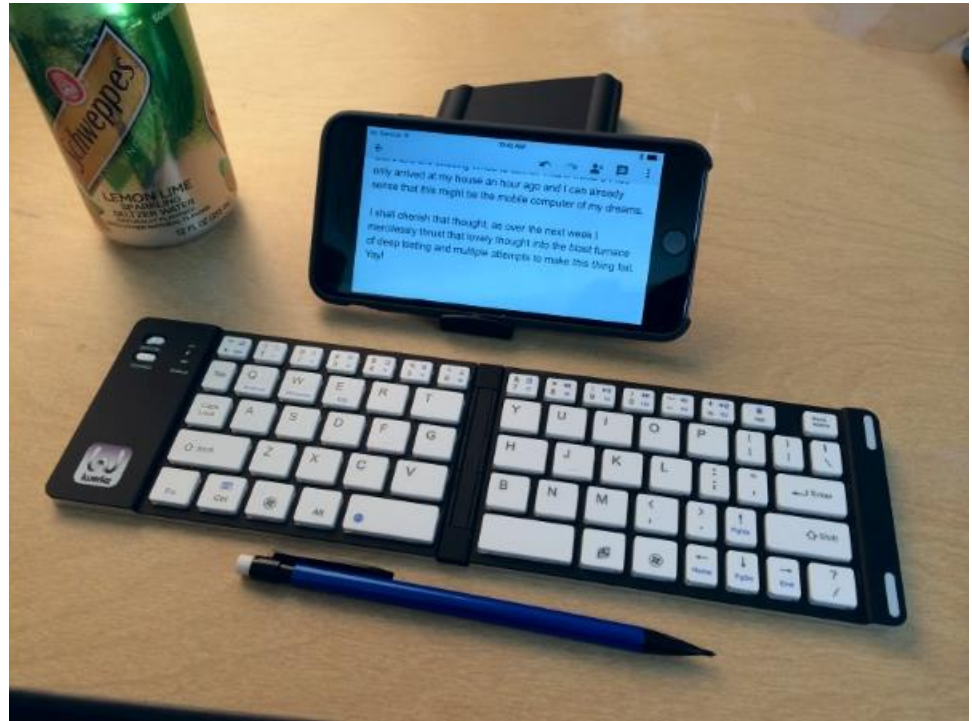
::Source::
Kamber (Sydney,
Australia)

Trends for Digital Media for 2016



Mobile Phone As Home Computer

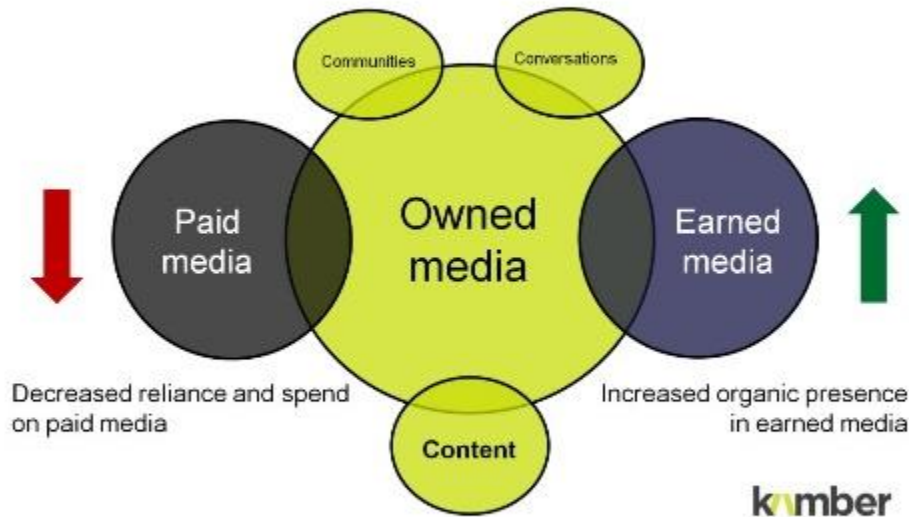
- Mobile Phone as PC
- Mobile phones are the new computer
- Designing website/social media for mobile phones is undoubtedly essential



voices.suntimes.com

Paid Media / Earned Media

The content marketing argument



::Source::
Kamber (Sydney,
Australia)

Creative-Digital Economy

- Placement: Its where young (and old) people are
- Price to invest: Minimal
- But..
 - Easy to build innovatively
 - Create HUGE profits



Future of Digital Media?



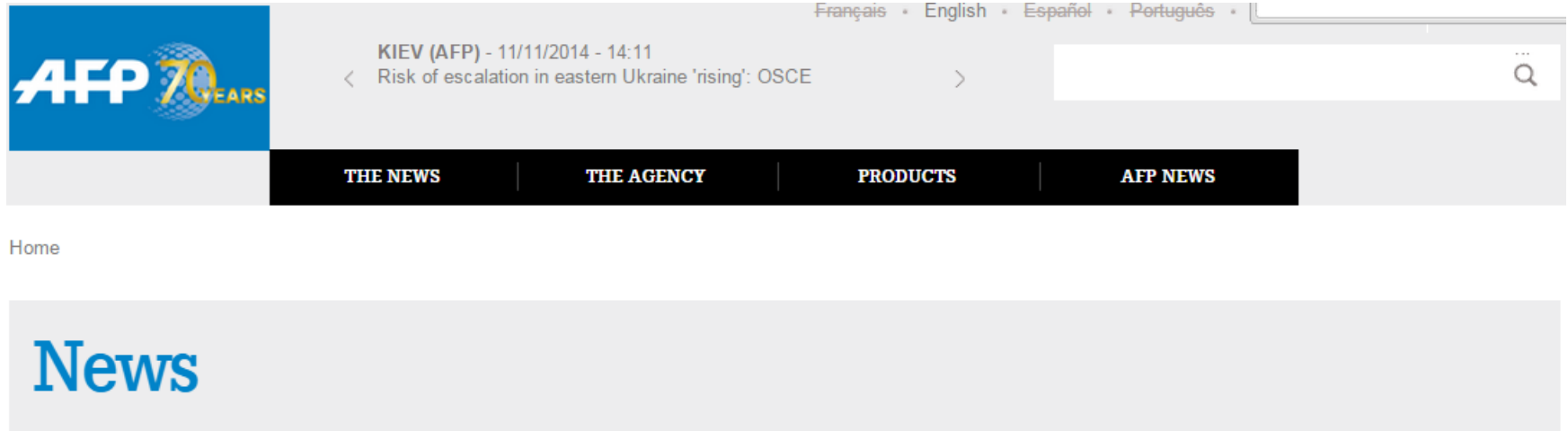
www.3dsg.com.au

- Big Data / SEO (Search Engine Optimization) / SEM (Search Engine Marketing)
- AI (Artificial Intelligence) / Deep Learning
- Reaching-Out-of-Computer
 - ▣ 3D-Display / 3D Television
 - ▣ MR (Mixed Reality) / AR (Augmented Reality) / VR (Virtual Reality)
 - ▣ New Digital Media

3D DISPLAY / 3D TELEVISION



3D Display / 3D Television



Japan firm showcases Bat-Signal of the future

20
OCT
2014

A free-floating image created by firing lasers into thin air was unveiled in Japan on Monday, offering the possibility one day of projecting messages into a cloudless sky, as seen in Batman.

And, like the Bat-Signal that alerted the Caped Crusader to trouble brewing in Gotham City, its inventors hope that it might eventually be used to warn people of danger such as an approaching tsunami.

The company behind the technology, Burton Inc., showed off rotating spirals, fluttering butterflies and the outline of an apple hovering a few metres over a van loaded with the system.

<http://www.afp.com/>

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3D Display / 3D Television

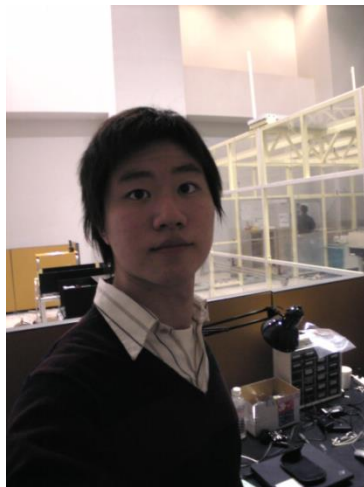
A free-floating image created by firing lasers into thin air



3D Display / 3D Television

2014 October

- 2008-Current
- Hideki Kimura
- National Museum of Emerging Science and Innovation (Miraikan), Tokyo



2009



<http://www.afp.com/>

3D Display / 3D Television



<http://www.afp.com/>

3D Display / 3D Television

- Miraikan is located in Odaiba, Tokyo



www.japan-guide.com

www.whitehouse.gov

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NEW VIEWPOINT ANALYSIS

Prof. Takeo Kanade,
Carnegie Mellon University (CMU), United States

Super Bowl



- Football fans treated to a unique new view of the action during playbacks
- CBS Television used a new technology
- Carnegie Mellon University computer vision expert, Prof. Takeo Kanade

Super Bowl

- “Eye Vision”
- Involves shooting multiple video images of a dynamic event, such as a football game, from multiple cameras placed at different angles
- The video streams from these cameras are combined by computer and the resulting images reach viewers in a format that will make them feel as if they are flying through the scenes they see.



Eye Vision from Prof. Takeo
Kanade, CMU

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The Matrix



□ “Virtualized Reality”

www.imdb.com

□ People will be able to customize the perspective from which they watch, for example, from that of a particular object.



AUGMENTED REALITY



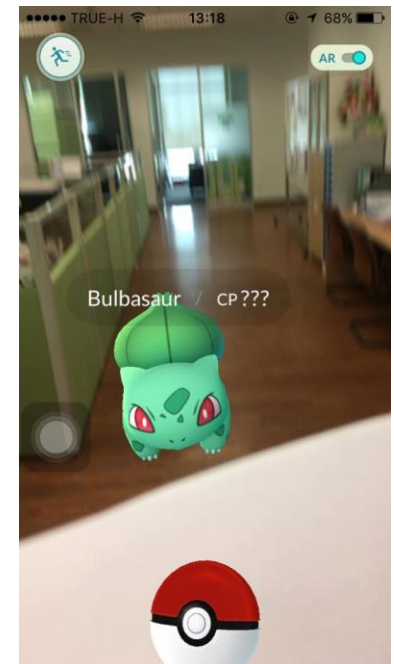
What is Augmented Reality (AR)?



- A combination of a real scene viewed by a user and a virtual scene generated by a computer that augments the scene with additional information.
- Generates a composite view for the user.

What is the Goal of AR?

- To enhance a person's performance and perception of the world
- But, what is the ultimate goal?



"Augmented reality in urban places: contested content and the duplicity of code."

Transactions of the Institute of British Geographers, 2012

The Ultimate Goal of AR

- Create a system such that no user CANNOT tell the difference between the real world and the virtual augmentation of it.



AR in Commerce Geek.com

Augmented Reality vs. Virtual Reality

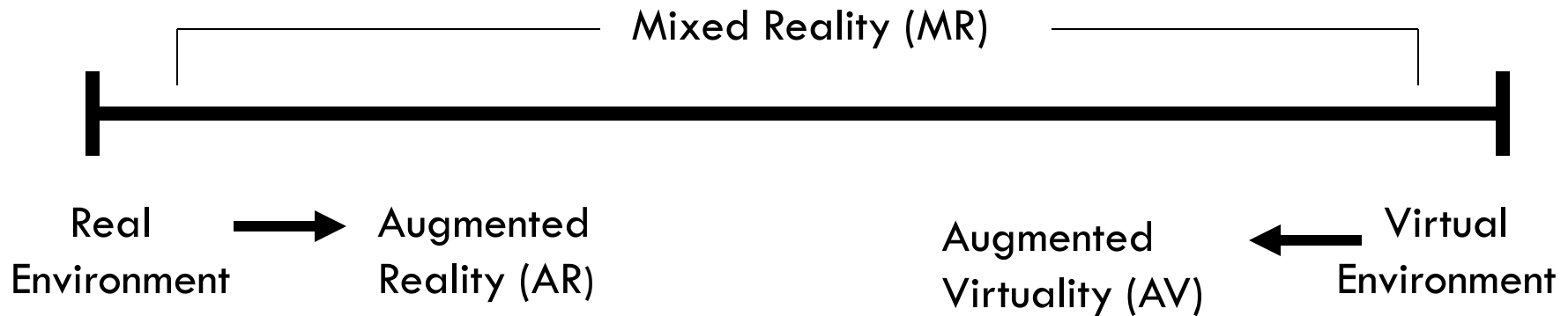
Augmented Reality

- System augments the real world scene
- User maintains a sense of presence in real world
- Needs a mechanism to combine virtual and real worlds

Virtual Reality:

- Totally immersive environment
- Visual senses are under control of system

Miligram's Reality-Virtuality Continuum



csis.pace.edu

Miligram coined the term “Augmented Virtuality” to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer generated data.

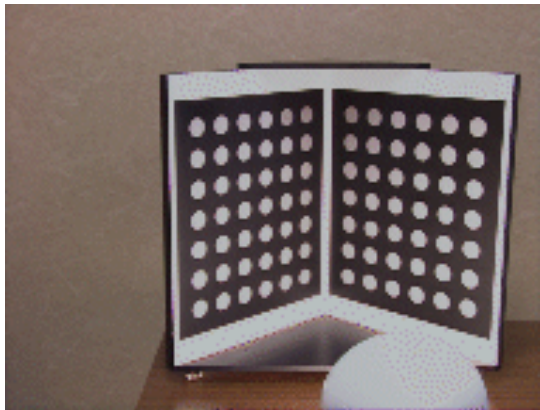
Combining the Real and Virtual Worlds

We need:

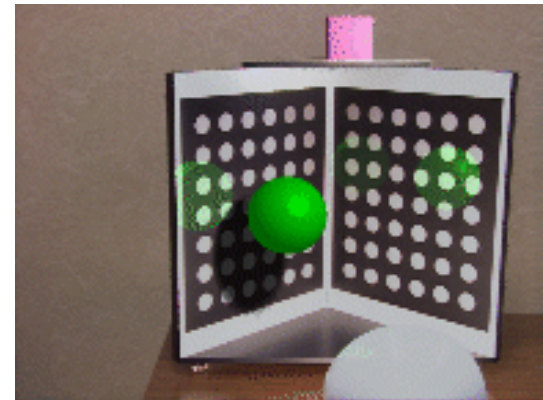
- ❑ Precise models
- ❑ Locations and optical properties of the viewer (or camera) and the display
- ❑ Calibration of all devices
- ❑ To combine all local coordinate systems centered on the devices and the objects in the scene in a global coordinate system

Combining the Real and Virtual Worlds (cont)

- Register models of all 3D objects of interest with their counterparts in the scene
- Track the objects over time when the user moves and interacts with the scene



real world



real world with virtual objects and inter-reflections and virtual shading

Mixed Reality

- Human pacman: A mobile wide-area entertainment system based on physical, social, and ubiquitous computing

Journal: Personal and Ubiquitous Computing

2004



Dr. Adrian Cheok,
National University of Singapore (NUS)
at Singapore

Broadcast in Discovery Channel

This image depicts an artist's conception of a mixed reality enhanced book. The reader is able to read the actual book, but simultaneously see three dimensional computer graphics co-located with, and in registration with the book.



www.sonycs1.co.jp

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Kerdvibulvech

Realistic Merging

Requires:

- ❑ Objects to behave in physically plausible manners when manipulated
- ❑ Occlusion
- ❑ Collision detection
- ❑ Shadows

****All of this requires a very detailed description of the physical scene**

Research Activities

- Develop methods to register the two distinct sets of images and keep them registered in real-time
 - ▣ New work in this area has started to use Computer Vision (CV) techniques
- Develop new display technologies for merging the two images

Performance Issues

Augmented Reality systems are expected:

- To run in real-time so that the user can move around freely in the environment
- Show a properly rendered augmented image

Therefore, two performance criteria are placed on the system:

- Update rate for generating the augmenting image
- Accuracy of the registration of the real and virtual image

Failures in Registration

Failures in registration due to:

- ▣ Noise

- Position and pose of camera with respect to the real scene
- Fluctuations of values while the system is running

- ▣ Time delays

- In calculating the camera position
- In calculating the correct alignment of the graphics camera

Applications

- ❑ Medical
- ❑ Military Training
- ❑ Hazard Detection
- ❑ Engineering Design
- ❑ Robotics and Telerobotics
- ❑ Training
- ❑ Sports
- ❑ Tourism
- ❑ Video conferencing

Applications (cont.)

- Entertainment, Computer Games
- Manufacturing, Maintenance, and Repair
- Consumer Design
- Audio
- Education
- Architecture
- Human welfare



Interactive Digital Multimedia Applications

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AUGMENTED REALITY APPLICATIONS



SIXTH SENSE



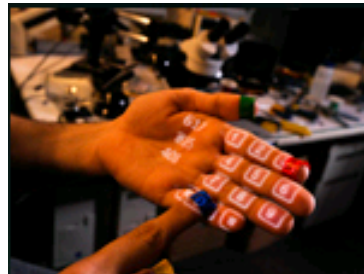
Sixth Sense



- A wearable gestural interface that augments the physical world around us with digital information
- Let us use natural hand gestures to interact with that information
- The SixthSense prototype is comprised of a pocket projector, a mirror and a camera.

Sixth Sense in 21st Century

TED
TALKS



Dr. Pattie Maes,
Massachusetts Institute of Technology
(MIT), United States

Broadcast in TED

2



Detail of Sixth Sense

- Sixth Sense allows people to use the internet without a screen or a keyboard
- It can turn any surface into a touch screen
- It is made with store-bought components
- *“The software program processes the video stream data captured by the camera and tracks the locations of the colored markers [that you wear on your fingers] using simple computer-vision techniques”..from MIT Media Lab*

Detail of Sixth Sense

- The device recognizes items in stores, and then gathers and projects information about products to the user.
- It also is able to provide quick signals to let users know what the best products are based on their needs.”
- Recognize articles in newspapers, retrieve the latest related stories or video from the Internet and then display them on pages for the user
- Say goodbye to your encyclopedia

Cost of Sixth Sense

- Only \$350 dollars, around 12,000 baht
- From an educator's stand point, this is minimum given the amount of money spent on textbooks, computer labs and LCD projectors.
- For educator's, Sixth Sense would be a way to put technology (literally) into each student's hands, because of its size and its cost.

AUGMENTED REALITY GUITAR



Augmented Reality Guitar



Dr. Chutisant Kerdvibulvech,
Keio University, Japan

Broadcast in Discovery Channel

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Assoc. Prof. Dr. Chutisant Kerdvibulvech

Thank you

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<https://www.facebook.com/chutisant.kerdvibulvech>